Topic: Prediction of house prices using machine learning

Tammanna khan

ABSTRACT

House price prediction is a beginner-friendly and widely used machine learning application in the real estate industry, assisting stakeholders in informed buying and selling decisions. This study focuses on predicting house prices using supervised machine learning techniques. Various regression models, including Linear Regression, Polynomial Regression, Decision Tree Regressor, and Random Forest Regressor, are explored. The dataset undergoes preprocessing steps such as handling missing values, feature scaling, normalization, and categorical feature encoding to enhance model performance. Model evaluation is carried out using Mean Absolute Error (MAE), Mean Squared Error (MSE), and R? score. Experimental results indicate that ensemble-based models outperform simple regression techniques, demonstrating improved accuracy and robustness in house price prediction.

INTRODUCTION

The dataset we are using:

	Id	Area	Bedrooms	Bathrooms	Floors	YearBuilt	Location	Condition	Garage	Price
0	1	1360	5	4	3	1970	Downtown	Excellent	No	149919
1	2	4272	5	4	3	1958	Downtown	Excellent	No	424998
2	3	3592	2	2	3	1938	Downtown	Good	No	266746
3	4	966	4	2	2	1902	Suburban	Fair	Yes	244020
4	5	4926	1	4	2	1975	Downtown	Fair	Yes	636056

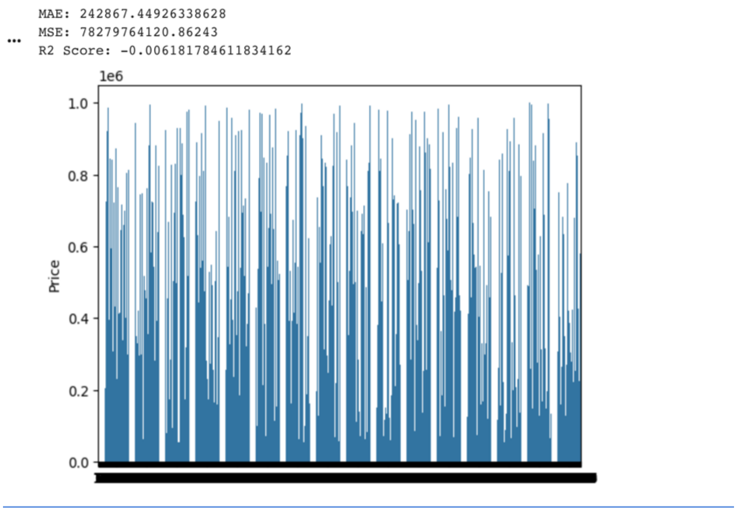
House price prediction is an important application of machine learning in the real estate sector, supporting data-driven decisions for buying and selling properties. The price of a house depends on multiple factors such as location, size, and available amenities, making manual valuation complex and inefficient. In this study, a supervised machine learning pipeline is implemented for house price prediction. The process includes data collection, data preprocessing (handling missing values, feature scaling, normalization, and categorical encoding), model training using regression algorithms, and performance evaluation. Regression models such as Linear Regression, Polynomial Regression, Decision Tree Regressor, and Random Forest Regressor are analyzed to determine their effectiveness. The models are evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2 score to identify the most accurate and reliable approach. This structured pipeline demonstrates the practical use of machine learning techniques for accurate house price prediction. The dataset is split into 80% training data and 20% testing data. One of the key challenges addressed in this study is the encoding of categorical features, as machine learning models cannot directly interpret data in categorical form. Therefore, appropriate encoding techniques are applied to convert categorical variables into numerical representations that can be effectively processed by the model.

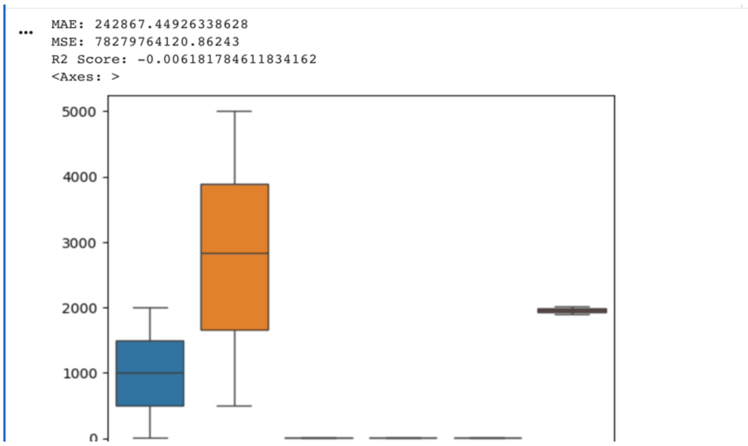
ADVANTAGES & DISADVANTAGES

Advantages: House price prediction helps buyers, sellers, and real estate professionals make informed decisions by providing data-driven price estimates. It saves time by automating the valuation process and helps analyze market trends using historical data. The system can handle large datasets efficiently and supports real estate businesses in setting competitive and realistic property prices.

Disadvantages: The accuracy of house price prediction models depends heavily on the quality and availability of data. Rapid changes in market conditions or economic factors can reduce prediction reliability. Some important features are difficult to quantify, and improper handling of categorical data may affect performance. Additionally, predictions are estimates and cannot completely replace professional property valuation.

RESULTS





CONCLUSION

This study utilizes a pipeline approach to simplify dataset processing and improve workflow efficiency. Logistic Regression was implemented as the learning model, and the accuracy score was used to evaluate the model's performance. The model is developed to benefit the real estate industry; therefore, all required datasets were used to achieve effective results. This work presents a beginner-friendly model with minimal complexity, making it easy to understand and implement. Future work can focus on incorporating advanced models and broader datasets to improve generalization and accuracy.

REFERENCE

[1] Harshita, Chinmay Kansal, Ayush Katiyar, and Adarsh Kr. Singh,

"House Price Prediction," Apr. 4, 2025.

[2] Anand G. Rawol, Dattatray V. Rogye, Sainath G. Rane, and Dr. Vinayak A. Bhradi,

"House Price Prediction Using Machine Learning," May 2021.

[3] [Kaggle](chatgpt://generic-entity?number=0),

"House Price Prediction Dataset,"

https://www.kaggle.com, accessed Apr. 4, 2025.